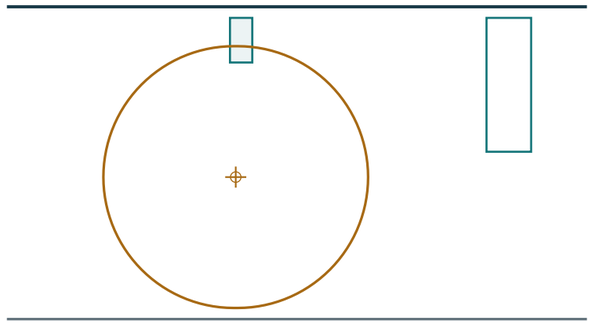
Wheel-clearance resolution trials: no option selected

E0 retains the conflict; E1 raises the full frame; E2 changes local members and seat-rail positions. Exact cylinder gaps and bounded motion results are distinguished below.

E0 / Retained 28 in floor

Conflict reference; D2 geometry

Floor 28 in / wheel D596 mm / P1 heave 25.4 mm



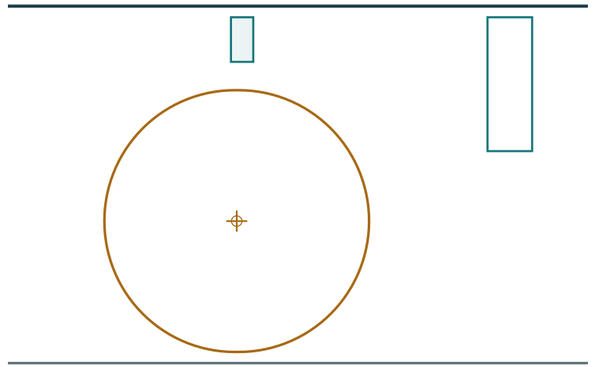
P1 exact minimum gap: -37.10 mm

110 members / 1,527.29 kg before joints

E1 / Raised 32 in floor

All steel moves up 4 in

Floor 32 in / wheel D596 mm / P1 heave 25.4 mm



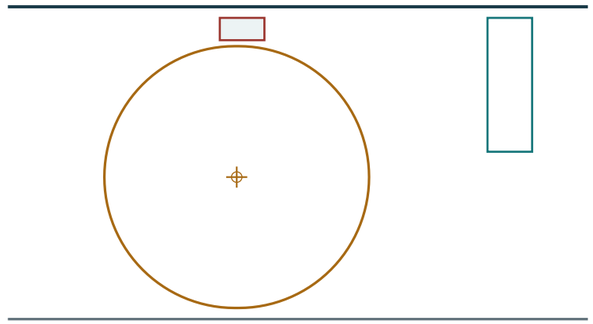
P1 exact minimum gap: +64.50 mm

110 members / 1,527.29 kg before joints

E2 / Local 2 in centre members

Inner seat rails move to y +/-26 in

Floor 28 in / wheel D596 mm / P1 heave 25.4 mm



P1 exact minimum gap: +13.70 mm

110 members / 1,525.44 kg before joints

G1 / ALL 110 MEMBERS CHECKED; SUPPLIER WHEEL / FLANGE AND MOTION STILL UNKNOWN

Profile / unselected sensitivity	E0	E1	E2
P0 /D596 /WB1165; no radial addition, heave or roll	-11.70 mm	+89.90 mm	+39.10 mm
P1 /D596 /WB1165; +25.4 mm heave; no radial addition or roll	-37.10 mm	+64.50 mm	+13.70 mm
P2 /BOUND: +25 mm radial, +25.4 mm heave, +/-2 deg roll	-81.47 mm	+20.13 mm	-30.67 mm
P3 /BOUND: +35 mm radial, +50.8 mm heave, +/-3 deg roll	-126.42 mm	-24.82 mm	-75.62 mm

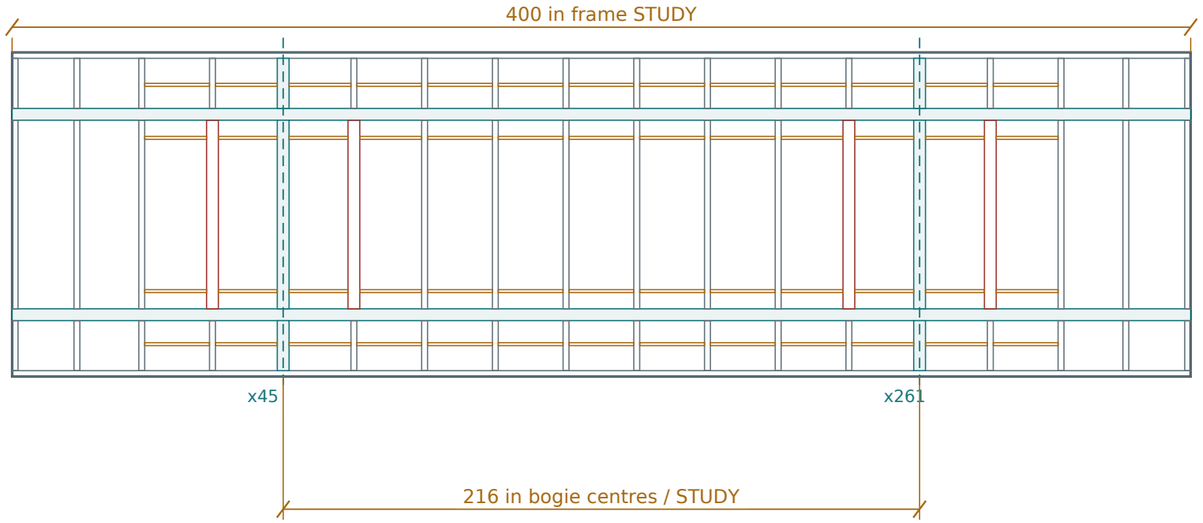
Negative exact-cylinder gap is proxy interference. Negative roll-bound gap is possible overlap, not confirmed collision. Positive gap at sampled states does not establish actual bogie fit or a required clearance. No hardware is selected.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

E2 load-path revision: shallow centres and relocated seat rails

All dimensions and member orientations are engineering studies. E2 changes local stiffness, seat-foot geometry and joints; it is not a fabrication-ready solution.

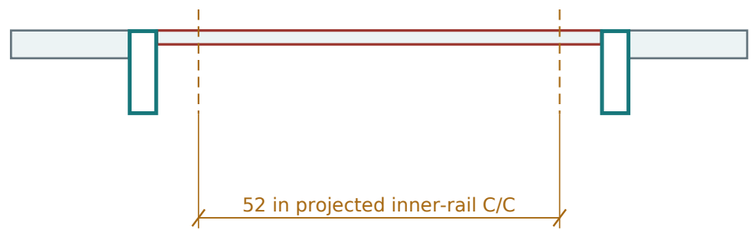
F1 / E2 FULL FRAME PLAN - CHANGED MEMBERS IN RED



Red central segments at x21 /69 /237 /285: 2 in deep x4 in along x.

Gold rails: inner y +/-26 in; outer y +/-44 in. Bench envelope unchanged.

F2 / SECTION AT ALTERED x21 - TOP FACES FLUSH



Central 4 x2 HSS is rotated to its weak bending axis.

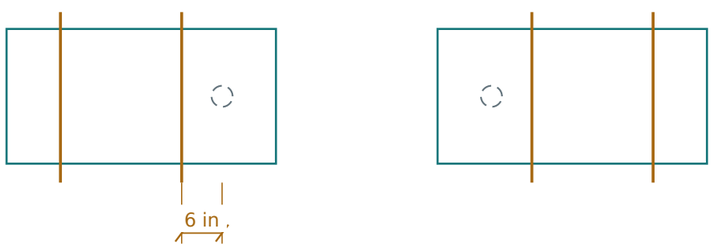
Outer arms stay 4 in deep; main girders and bolsters stay12 in.

Steel top z27.0039 in; central underside z25.0039 in.

Gold = adjacent seat-rail centrelines projected; joints HELD.

Weak I=0.898 vs2.65 in4; S=0.898 vs1.33 in3.

F3 / SEAT BASE INTERFACE CHANGE - PLAN RESERVATIONS



Dashed circles = prior inner lines; actual feet / anchors / joints HELD.

F4 / LOAD TRANSFER TO PROVE

Seat -> actual base and anchors -> moved rails -> shallow centre members -> main girders.

Engineer strength, stiffness, local wall response, fatigue and the changed joints together. Touching envelopes are not connections.

F5 / RETAINED SUPPORT RESPONSIBILITIES

The x237 member also supports a panel seam. A wheel-clearance change must preserve the floor joint, seat rail and frame load paths.

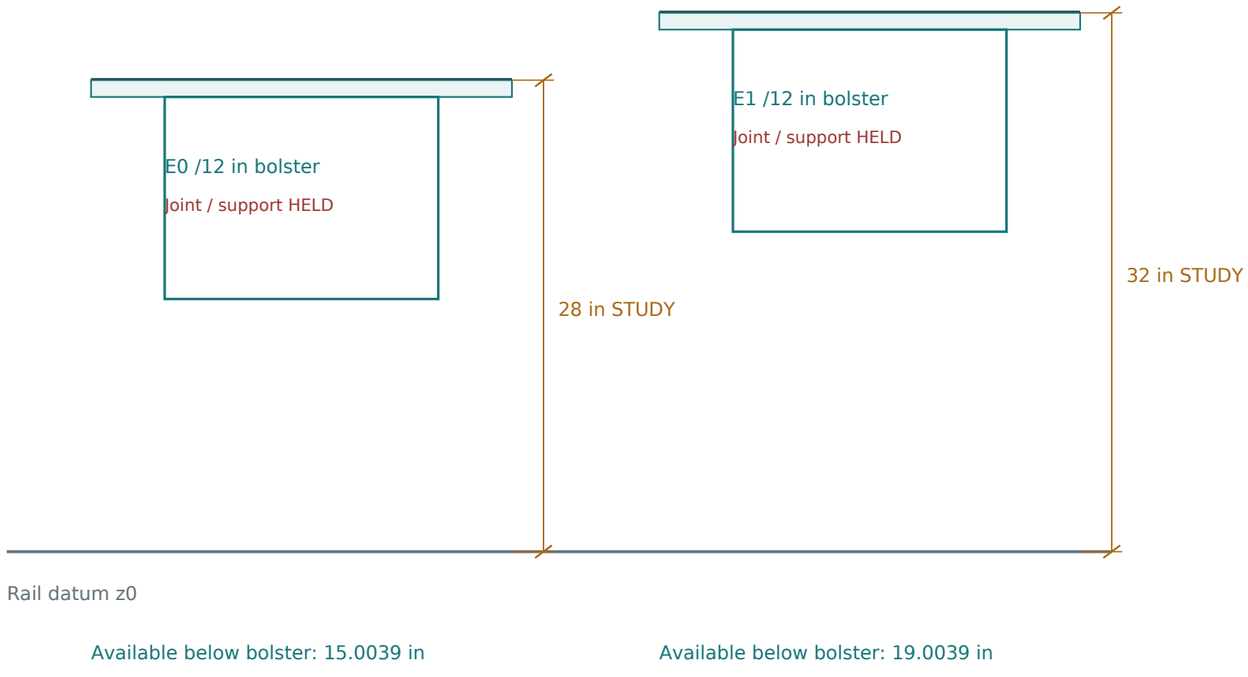
No cutting, notching, welding or procurement is released by this sheet. Confirm the entire 3D clearance envelope first.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Floor height: clearance, boarding and body consequences

Compare the retained 28 in floor with a 32 in sensitivity. Real support height, travel, interface stack, centre of gravity and boarding design remain supplier / engineer returns.

R1 / RAIL-DATUM ELEVATION - FULL FRAME RAISE E1



E1 raises all steel, body and floor datum by 4 in; nominal frame mass is unchanged.

R2 / REQUIRED HEIGHT CONTRACT

OEM support Hs + joint J + motion C
<= floor F -25.30/25.4 -12 [in]

Each value must state load condition, wheel wear, suspension position, units and measurement datum. Do not substitute a catalog overall height for the actual body support plane.

Interface	E0 /E2	E1
Finished floor	28.0000 in	32.0000 in
Steel top	27.0039 in	31.0039 in
12 in bolster bottom	15.0039 in	19.0039 in
Actual Hs /J /C	HELD	HELD

E2 retains the lower floor and changes four centre members and inner seat rails. Its local shallow member does not raise the 12 in bolster, main girders or actual bogie support allowance.

R3 / COORDINATE BEFORE CHOOSING A FLOOR

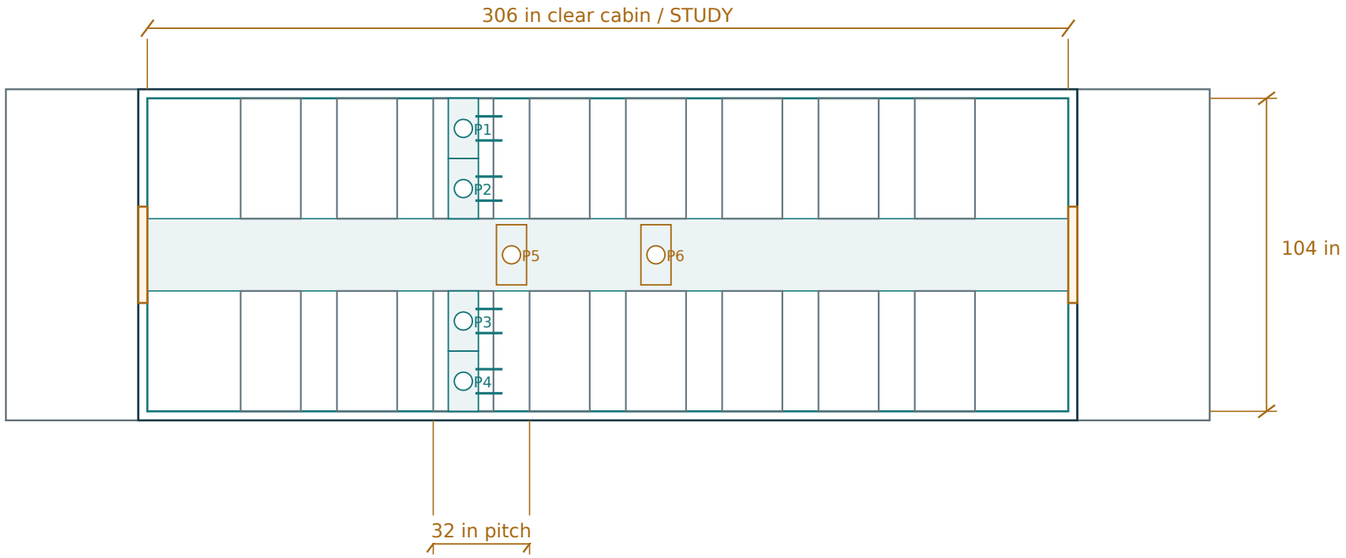
The +4 in option changes steps, accessible boarding, overall height and centre of gravity. Preserve clear cabin headroom; recheck station interfaces, roof equipment, body roll and the full route envelope. No station height is fixed by this sheet.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Regular coach mockup: measure real adults before fixing the cabin

Ground-level tape and rated chairs only. Six displayed scale proxies do not prove population fit, egress, comfort or standing capacity; actual participant results remain unperformed.

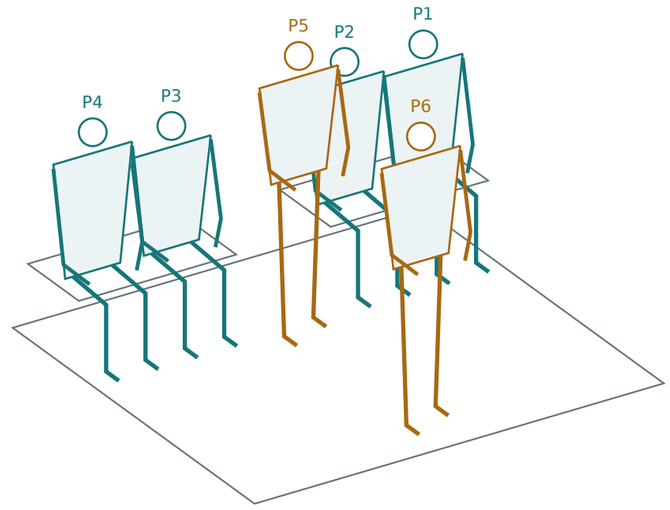
M1 / FULL-SIZE TAPE PLAN - RETAINED 306 x104 in CLEAR CABIN



Display: P1 through P4 seated; P5 and P6 standing. Main aisle 24 in; benches 40 in each.

Mark every row; use six rated chairs in the active two-row trial, not a load-bearing prop.

M2 / SIX ADULTS AT THE SAME GEOMETRIC SCALE



Illustrative heights 65-74 in; seat target 18 in; head / limbs are proxies.

Seat planes are targets only; actual rated-chair footprint must be recorded.

M3 / CHANGE ONE DIMENSION AT A TIME

Trial	Aisle /bench width	Per-adult arithmetic
RA24	24 /40 in	20 in per bench half
RA28	28 /38 in	19 in per bench half
RA32	32 /36 in	18 in per bench half

RA28 / RA32: paired 19.2 in chairs do not fit. Use a suitable rated fixture.

M4 / RECORD THE SPACE THAT IS ACTUALLY USABLE

Use varied adults in winter clothing. Measure knees, feet, aisle intrusion, passing, getting up, carrying belongings and both end-door approaches.

Eight-row pitch trials 32 / 34 / 36 in leave rear end zones 31 / 17 / 3 in before occupied bodies and door hardware. Do not adopt a longer pitch from empty-seat arithmetic.

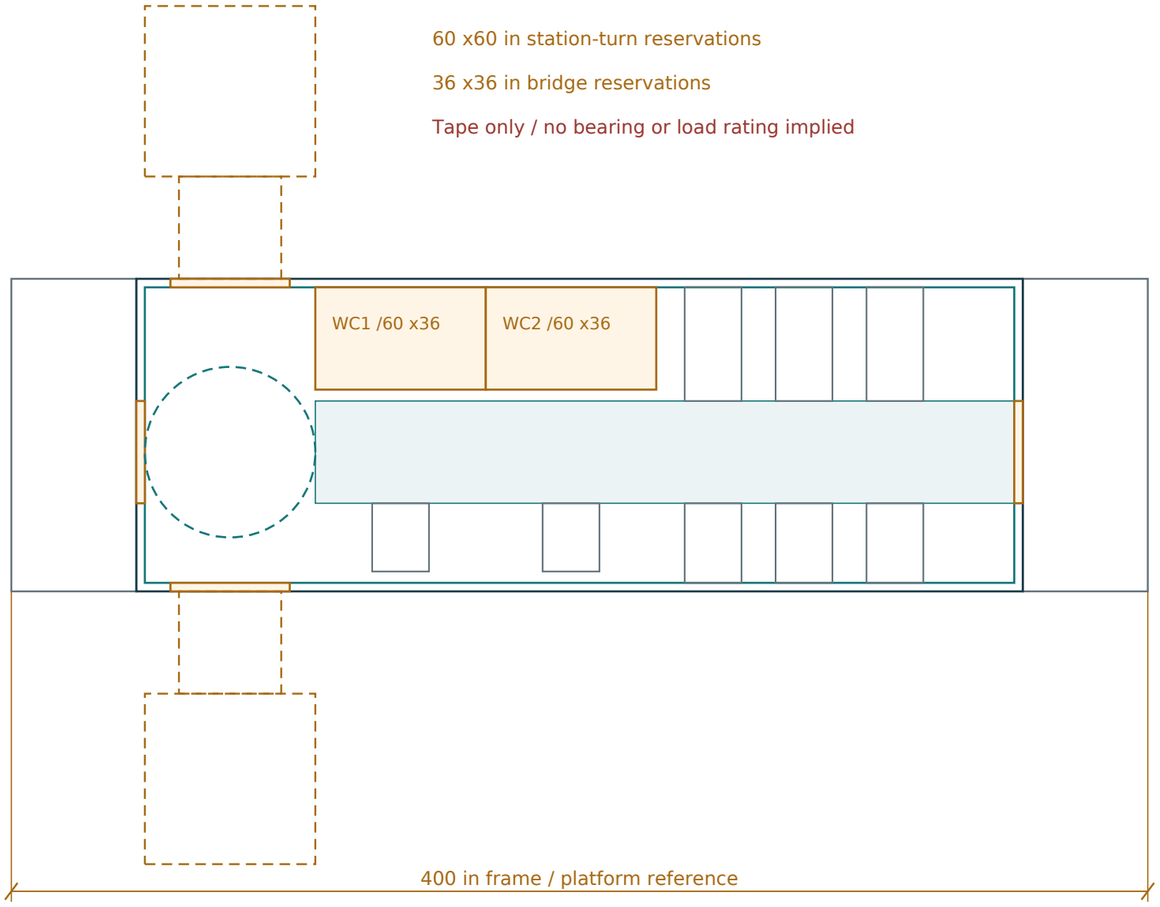
Reference chair depth 21.4 in leaves 10.6 in footprint gap at 32 in pitch. Check the 78 in end-door target and full cabin headroom.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Accessible mockup: actual devices, companions and station approaches

All marks are on the existing floor. No improvised ramp, lift or raised occupied deck is authorized. Requirements and vehicle classification remain subject to responsible review.

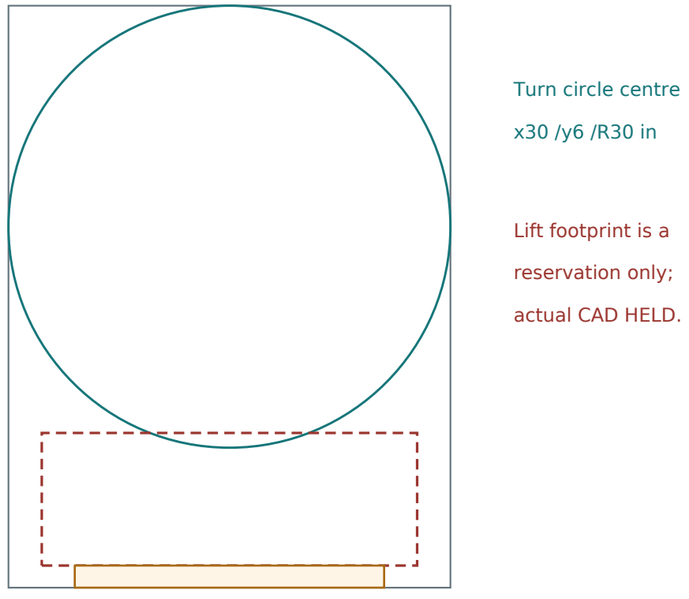
A1 / FULL-SIZE FLOOR TAPE - BOTH BOARDING SIDES



Two 60 x 36 in bays; route 36 in; side door 42 in; companion locations retained.

Use real mobility devices and assistants. Record turning, feet and luggage sweep.

A2 / OPTIONAL LIFT RESERVATION / FRONT-TURN CONFLICT



The retained circle reaches y36; optional reservation begins y34.

This is 2 in of nominal overlap before handrails / occupied hardware.

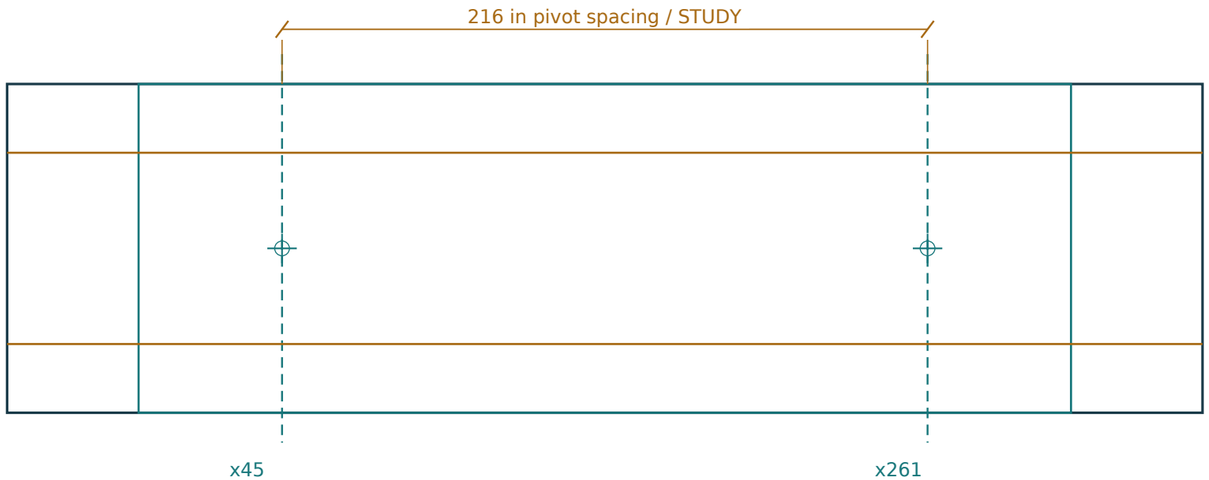
A shifted circle and wider side-door trial may be taped separately. They do not select an onboard lift or establish boarding or evacuation compliance. Qualify both normal stations and independent stalled-train rescue.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Supplier return drawing: one coordinate and load contract

Blank OEM fields are intentional. Return configured manufacturer/model, drawing revision, load condition, limits and native CAD; gauge agreement alone does not establish compatibility.

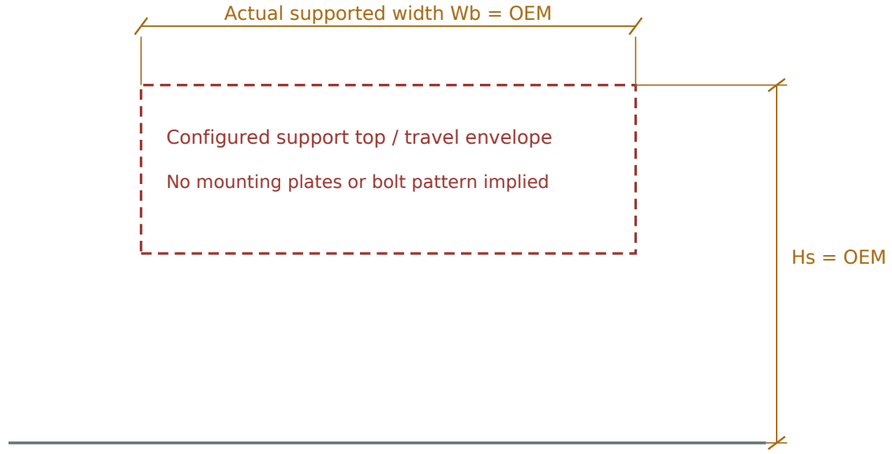
I1 / COACH DATUM PLAN - KNOWN STUDY GEOMETRY



Origin: x0 front cabin inside wall; y0 vehicle centreline; z0 top of rail.

Gold: main-girder inner faces y +/-32 in; not an approved bogie bay.

I2 / RETURN CROSS-SECTION - SYMBOLIC ONLY



Return new / worn wheel and full suspension extrema, not a single pose.

I3 / REQUIRED RETURN FIELDS

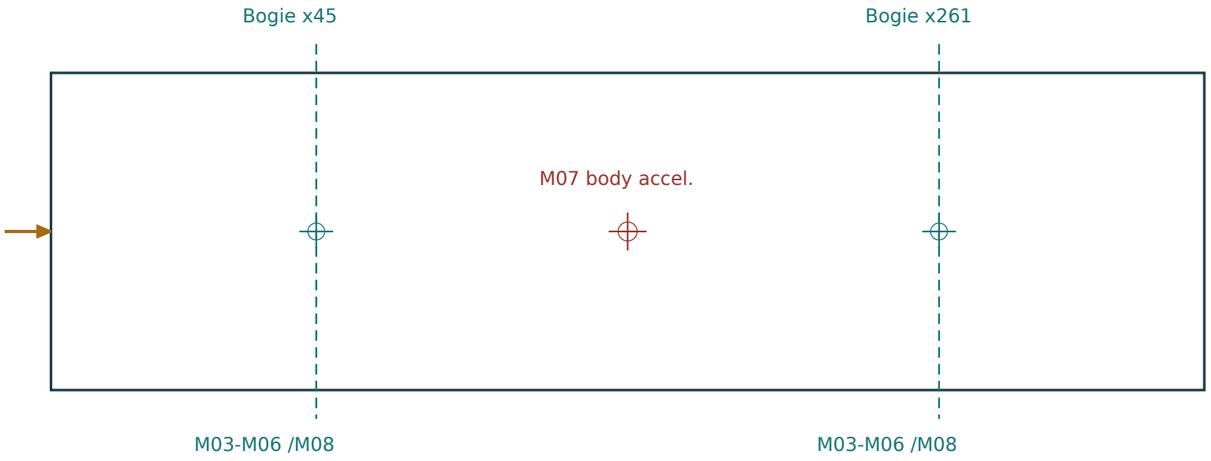
ID	Return in the shared datum	Source / units / load condition / revision
RG-1	Wheel / rail profile; gauge, back-to-back, check dimensions	
RG-2	Support height, pivot / side-bearer interfaces and travel	
RG-3	Wheelbase; full brake / axlebox / suspension swept CAD	
RG-4	Rated load spectrum, braking / roll moments and maintenance removal	

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Prototype measurements: common time, separate physical evidence

Proposed instrumentation only. Actual mounts, pressure ports, force transducers, calibration, test loads, limits and isolation/restraint procedures require specialist release.

T1 / COACH MEASUREMENT ZONES - NO SENSOR MOUNTS RELEASED



M09 joint-force / stroke assembly requires engineered installation.

M02 independent ground speed / distance; M01 common time base.

T2 / MEASUREMENT INDEX

ID	Recorded quantity
M01-M02	Events / common clock; independent speed and distance
M03	Individual axle speeds; all four coach axles
M04-M05	Working and auxiliary-release pressures, each bogie
M06-M08	Actuator travel; body acceleration; disc / pad temperatures
M09	Engineered joint force and stroke measurement
M10-M11	Brake DC voltage/current; hotel three-phase power
M12-M13	Cabin / ambient T, RH, airflow; emergency lighting
M16	Calibrated wheel-load and total-mass measurements

Pressure alone is not brake force. Derive force/torque only from a documented actuator convention and geometry; correlate wheel speeds, temperature and acceleration on one time base.

T3 / SYNCHRONIZED RECORD - FUNCTIONAL DATA MAP

Mechanical / hydraulic channels
Speeds, travel, pressure, temperature

→

M01 synchronized event record
Time, test state, applied fault and reset

←

Electrical / environment channels
DC, AC power, cabin / ambient condition

Independent witness record
Calibration / photos / disposition

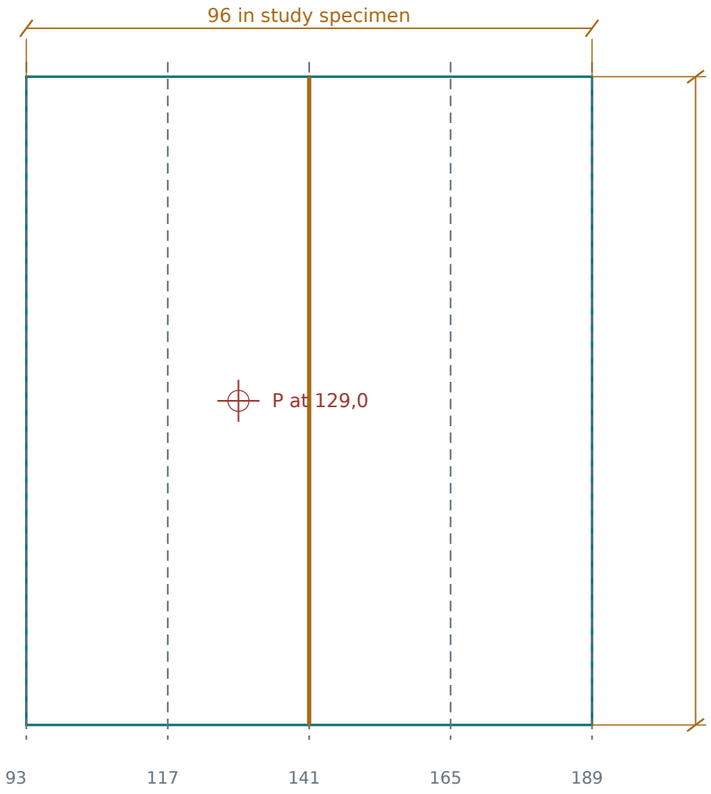
Record baseline, controlled application, fault response, recovery and residual condition. Separate commanded values from measured values. Testability and emergency access must be maintained throughout the temporary installation.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Floor specimen and fault evidence: define what must be measured

These are test-development diagrams. Applied loads, patch area, cycles, safe restraint, instrument ranges and acceptance criteria are intentionally unassigned pending engineer / OEM procedures.

T4 / TWO FULL-WIDTH FLOOR PANELS / SOURCE COORDINATES



M14: deflection / independent stable datum

M15: relative seam movement

M17: simultaneous support displacement

Support stations: 93 / 117 / 141 / 165 / 189

Use M14 and M17 to distinguish total and panel-relative deflection. Return actual substrate, edge closure, supports, adhesive / fasteners, finish and conditioning; test the installed load-introduction details.

Cross denotes patch centre only. Patch size, proof load and repeat cycles remain HELD; there is no test ballast schedule on this sheet.

T5 / FAULT-EVIDENCE STATES / UNPERFORMED TESTS

Test	Stage	Record
E03	OEM bench	Force / pressure / travel mapping
E04	Secured coach	Application, inhibit, reboot / reset
E05	Secured coach	Separation and communication faults
E10	Electrical setup	Source transfer and protection
E12	Released setup	Blackout / E-stop / emergency light

A command to brake is not evidence of brake force. Preserve time-correlated measured values before, during and after each fault and restoration. Introduce faults only as defined by the released procedure.

Required record fields: configuration revision, approved protocol and limits, instrument calibration, test director, witnesses, actual result, nonconformance disposition and release signature. All remain blank until performed.

T6 / NO SILENT PASS OR AUTOMATIC REPLICATION

The accepted first coach is the configuration reference for the remaining vehicles. Any changed frame, wheelset, brake, seating anchor, power package or software requires a documented effect review and affected retests. A first-coach test does not release the full train: complete-consist brake balance, response, emergency operation and evacuation need their own evidence.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release